Age of the Universe

Exam code: 0625 & 0972

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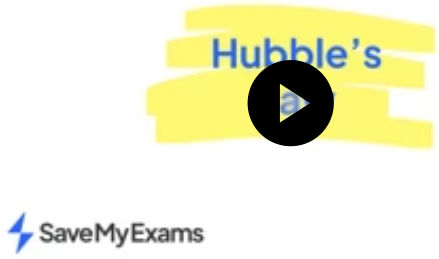
Updated on 1 October 2024



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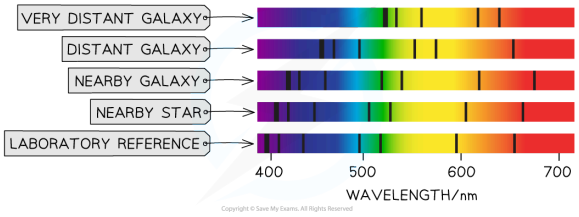
The Hubble constant

Extended tier only

- When Edwin Hubble looked at the absorption spectra of distant galaxies, he determined a relationship between the **speed** of a galaxy and its **distance** from Earth

Comparing redshifts of galaxies



Hubble discovered that all galaxies show redshift, but the galaxies that are further away show a greater increase in redshift

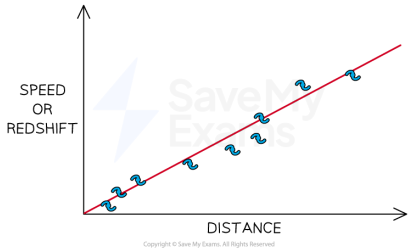
- This is **Hubble's law**, which states

The speed of recession is proportional to the distance of the galaxy away from Earth

- 'Recession' speed means the speed at which something is moving away

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Relationship between redshift and galaxy distance



Hubble's law tells us the greater the distance to a galaxy, the greater the redshift, or the speed it moves away from Earth

- Hubble's law can be expressed as an equation:

$$H_0 = \frac{v}{d}$$

- Where:
  - H_0 = Hubble constant (per second)
  - v = recessional velocity of an object, the velocity of an object moving **away** from an observer (km/s)
  - d = distance between the object and the Earth (km)
- From this equation, the Hubble Constant H_0 can be defined as:

The ratio of the speed at which the galaxy is moving away from the Earth, to its distance from the Earth

- The accepted value of the Hubble constant is H_0 = **2.2 × 10^{-18} per second**

Examiner Tips and Tricks

Make sure to learn the currently accepted value of the Hubble constant.

You will be expected to know that the current estimate for H_0 is 2.2 × 10^{-18} per second

Measuring recession speed & distance

Extended tier only

- The Hubble constant H_0 can be determined from measurements of:
  - redshift of the light emitted by a galaxy
  - the brightness of supernovae in the galaxy

- The speed of recession v of a galaxy (i.e. how fast it is moving away from the Earth) can be found from the change in wavelength of the galaxy's starlight due to redshift

Measuring distance using supernovae

- The distance d to a galaxy can be determined using the brightness of a supernova in that galaxy
- **Supernovae** are exploding stars
  - Certain types of supernovae have the same peak level of brightness (**absolute magnitude**), meaning they can be used as **standard candles**
  - These supernovae are so bright that they can be used for measuring distances to the most distant galaxies

Age of the Universe

Extended tier only

- Hubble's law can be rearranged to give the expression:

$$\frac{1}{H_0} = \frac{d}{v}$$

- Since time is equal to distance divided by speed, the term $\frac{1}{H_0}$ represents an estimate of the **age of the Universe**
- Hubble's law provides further **evidence** for the Big Bang
- **It shows that the Universe has been expanding since the beginning of time**
  - If we looked at time in reverse, we would see galaxies were closer together in the past
  - This suggests that the Universe must have originated from a single point and has been expanding outwards ever since

Hubble's law graph

- Using measurements from galactic redshift and brightness of supernovae, a graph of recession velocity against distance can be plotted

Graph of galaxy recession velocity against distance

A key aspect of Hubble's law is that the furthest galaxies appear to move away the fastest

- The **gradient** of the graph represents the age of the Universe
  - When the distance equals zero, this represents all the matter in the Universe being at a single point
  - This is the singularity that occurred at the moment of the Big Bang
- Astronomers have used this formula to estimate the age of the Universe is about 13.7 billion years

Worked Example

A distant galaxy is 20 light-years away from Earth.

Use Hubble's law to determine the velocity at which the galaxy moves away from Earth.

Answer:

Step 1: List the known quantities

- Distance to the galaxy, d = 20 light-years
- 1 light year = 9.5×10^{15} m
- Hubble constant, H_0 = 2.2×10^{-18} per second

Step 2: Convert 20 light-years to m

- d = 20 ly = $20 \times (9.5 \times 10^{15}) = 1.9 \times 10^{17}$ m

Step 3: Substitute values into Hubble's Law

$$v = H_0 d$$

$$v = (2.2 \times 10^{-18}) \times (1.9 \times 10^{17}) = 0.418 \text{ m/s}$$

- The galaxy moves away from Earth at a velocity of 0.42 m/s

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